

Supporting Data Availability & Author Declarations Statement

Target Journal: *IEEE Transactions on Systems, Man, and Cybernetics: Systems* (IEEE TSMC: Systems)

Manuscript Title: Value Chain Physics: Industrial Validation and Formal Proof of Open Complex Giant Systems Based on Non-IID Theory

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1. Data Availability Statement

The datasets, topological network structures, and computational benchmark scenarios analyzed in this study originate from 18 years of industrial physical practice in global supply chain systems (Lenovo Group IPS System) and open-source complexity models.

- **Public Repository:** All core topological data structures, algorithm pseudo-code, mathematical proof verifications, and benchmark models supporting the conclusions of this paper are deposited and publicly accessible in the open-source GitHub repository:

<https://github.com/GritMeng/Value-Chain-Physics>

- **Preprints and Extended Documentation:** SSRN Working Papers (Abstract IDs: 7451920, 7442100, 7251098).
- **Proprietary Data:** Any specific confidential commercial transactions have been fully anonymized into normalized topological graph matrices in the open repository to ensure complete reproducibility without violating commercial non-disclosure obligations.

2. Code & Software Availability

- The Intelligent Planning and Control (IPC) engine algorithms and high-dimensional phase space topological solvers are open-sourced at <https://github.com/GritMeng/Value-Chain-Physics>.
 - The algorithm requires standard Python 3.9+ environment with NumPy / SciPy packages. No proprietary third-party software licenses are required to execute the verification scripts.
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3. Funding Statement

This research received no grant, financial support, or sponsorship from funding agencies in the public, commercial, or non-profit sectors. All theoretical research, mathematical proofs, and software development were conducted independently by the author using private computational resources.

4. Conflicts of Interest / Competing Interests

The author declares no financial, personal, or professional competing interests, affiliations, or financial holdings that could influence or appear to influence the objectivity of the research presented in this manuscript.

5. Ethical Compliance and Human/Animal Rights

This research does not involve human participants, clinical trials, personal identifiable data, or animal subjects. Ethical approval and informed consent were not required for this purely theoretical and algorithmic study.

6. Consent for Publication

The author consents to the publication of this manuscript, title page metadata, and supporting statements in *IEEE Transactions on Systems, Man, and Cybernetics: Systems*.